Aman Keshav Patre

Data Analyst — Data Scientist — Machine Learning Engineer

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Professional Summary

Computer Engineering graduate (CGPA: 9.13, June 2026) with industry experience as a Data Science Intern at UPL Limited. Skilled in Python, SQL, Power BI, and machine learning with a track record of building end-to-end ML pipelines, customer analytics platforms, supply chain models, and IoT-integrated dashboards. Actively seeking roles in **data analytics, data science, or ML engineering**.

Experience

Data Science Intern **July 2025 – September 2025**

UPL Limited, Gujarat — Energy Optimization & Video Analytics

- Cleaned and analyzed 3+ industrial IoT sensor datasets using Python (Pandas, NumPy), identifying 200+ anomalous records across shift-level production logs to surface actionable operational inefficiencies.
- Built optimization models across 6 operational parameters, improving plant energy efficiency KPIs and reducing manual reporting effort for weekly cross-departmental reviews.
- Developed 3 interactive Power BI dashboards tracking real-time energy consumption metrics, adopted by 2 engineering departments for daily operational decision-making.
- Annotated 1,000+ video frames with structured quality criteria to build training datasets for a computer vision analytics pipeline, maintaining defect-log documentation throughout.

Projects

Customer Analytics Suite — Segmentation & Churn Prediction Python, K-Means, PCA, SVM, Streamlit

Segmentation Demo — *Churn Demo* — *Churn Repo*

- Applied K-Means (Elbow Method) on 2,200+ customer records to identify $K = 6$ behavioral segments; reduced 7 features to 2 via PCA and deployed a Streamlit dashboard with `joblib` model caching and churn-risk UI alerting.
- Extended the pipeline with an SVM churn classifier achieving **94.5% accuracy** (Precision: 0.94, Recall: 0.95) via 5-fold cross-validation grid search; supports real-time single-account diagnosis and batch CSV risk auditing.

Global Supply Chain Fulfillment Analytics & Delay Prediction Python, Random Forest, SMOTE, EDA

Technical Report

- Diagnosed a 54.7% delivery failure rate across 180,519 global transactions (2015–2018), quantifying **$2.1M in eroded profitability** from contractual penalties and emergency carrier rerouting.
- Trained a Random Forest Classifier with SMOTE oversampling and frequency encoding achieving **82% accuracy** and **86% precision** on late-delivery flagging; ROC-AUC of 0.91 enables point-of-sale risk alerting before shipment.

UIDAI Biometric Demand Intelligence Dashboard Python, Streamlit, Predictive Analytics, Time-Series

Live Demo — UIDAI National Hackathon

- Processed **1.8M+ data points** to generate 12 regional Aadhaar biometric enrollment forecasts, enabling data-driven allocation of devices, operators, and facilities for national identity infrastructure.
- Built automated operational risk modeling with demographic filtering (age cohorts 5–17 and 18+) and state-level KPI tracking to detect high-risk enrollment months across 12+ regions.

Smart Anna — Intelligent Cafeteria Management System Python, IoT, Supabase (PostgreSQL), Vercel

Live Demo

- Developed an IoT-enabled full-stack platform with real-time occupancy dashboards (5+ zone sensors) and Supabase (PostgreSQL) backend handling 200+ daily transactions with normalized relational schemas.
- Deployed digital ordering, revenue analytics, and operational KPI dashboards on Vercel; integrated menu management and order tracking modules.

AgriYield Pro — Crop Yield Prediction Platform Python, Streamlit, Random Forest, Scikit-learn

- Trained a Random Forest pipeline predicting crop yield (hg/ha) across 10+ crop types on 5,000+ test samples; covered the full ML lifecycle from preprocessing to deployment.
- Built an interactive Streamlit dashboard with dynamic inputs, prediction history, CSV export, and feature importance visualization.

EduPredict Pro — Student Performance Prediction System Python, Scikit-learn, Streamlit, Plotly

Live Demo

- Achieved **88% prediction accuracy** with a Linear Regression pipeline (One-Hot Encoding + StandardScaler) on 7 student features across a multi-demographic dataset.
- Delivered 3 modules — single-prediction UI, batch CSV processing, and a Scenario Simulator — with Plotly visualizations and integrated email reporting.

Languages & Technologies

Languages	Python, SQL, MySQL
ML / AI	Scikit-learn, XGBoost, SVM, K-Means Clustering, PCA, Random Forest, Linear Regression, SMOTE, TensorFlow, PyTorch, Pandas, NumPy, SciPy, Seaborn, OpenCV, Matplotlib, Plotly
Analytics	Power BI, Streamlit, Excel (Advanced), EDA, Statistical Analysis, RFM Analysis, Cohort Analysis, A/B Testing
NLP	Text Preprocessing, TF-IDF Vectorization, Classification Pipelines, LLM Concepts
Engineering	Feature Engineering, Predictive Modeling, Unsupervised & Supervised Learning, ETL Pipelines, Optimization Modeling, Computer Vision
Cloud/Tools	Supabase (PostgreSQL), Vercel, Google Cloud, Git, GitHub, Jupyter Notebook, Google Colab, joblib, Dill, Streamlit
Coursework	Machine Learning, DBMS, Data Mining, Statistics, Artificial Intelligence

Education

B.E. Computer Engineering **June 2026**
UPL University of Sustainable Technology, Gujarat
CGPA: 9.13

Certifications

- Developing Data Science Projects with Google Colab — Udemy (NLP, classification, end-to-end ML workflows)
- Data Analysis Using Python
- Deloitte Australia Data Analytics Job Simulation — Advanced analytics, business insight reporting